

Weekly Lesson Plan – Week 7 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon Week of: September 21, 2026 Course: Engineering Design & Problem Solving (InSPIRESS)				
TEKS / ELPS:				
TEKS: §127.406(6) implement a design process to develop a multi-system product or solution ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G				
Aim of the Lesson / Objective: <i>Statement of what you want students to do (skill) and/or know (knowledge). Students will...</i>				
MONDAY SEP 21	TUESDAY SEP 22	WEDNESDAY SEP 23	THURSDAY SEP 24	FRIDAY SEP 25
Students will complete the toy block model using shell, extrude and fillet, and save it to their own work folder.	Students will hollow a solid model using the shell feature and add internal support columns with a sketch and extrude.	Students will apply fillets to finish the toy block and edit an existing feature from the timeline instead of rebuilding it.	Students will complete the toy block, name the file to the class standard, and explain their build to the teacher at check-in.	Students will finish and save their toy block file and define the CAD terms used in the build in their own words.
Employed Language Domain(s) Objective: <i>How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...</i>				
Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members’ ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.				
Assessment of Content Objective: <i>Instrument used to determine mastery. Ex: Exit Ticket.</i>				
Students will be assessed by written, group, or oral responses throughout the class period.				
Set (Bell Work): <i>A motivating action that engages students in the objective – question, fact, activity. Essential Question:</i>				
MONDAY SEP 21	TUESDAY SEP 22	WEDNESDAY SEP 23	THURSDAY SEP 24	FRIDAY SEP 25
A company can ship a product one month early if it skips a round of testing. Nothing has failed so far. Do you ship early?	A soda bottle, a cup, and a wheel are all round. What do they have in common in how they could be made?	Your design works, but a coworker’s version is simpler and cheaper. Your name is already on yours. What do you do?	Glass, aluminum, or plastic for a soda bottle. Pick one and defend it.	What are today’s key ideas in: Half day (conferences): save, check, catch up?
Methodology / Steps: <i>What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...</i>				
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Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): ship a month early by skipping a round of testing? (10 min) Step 3: Finish Day 1: shell through the bottom, inside columns, fillet (setup first if not signed in) (30 min) Step 4: Save over the same file (Ctrl+S) (3 min) Step 5: Finished? Help someone at your table finish theirs (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: a bottle, a cup and a wheel are all round - how could all three be made? (10 min) Step 3: Shell through the bottom face, then sketch and extrude the inside columns (30 min) Step 4: Save over the same file (Ctrl+S) (3 min) Step 5: Spin the model: studs on top, hollow underneath (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): a coworker’s version is simpler and cheaper than yours - what do you do? (10 min) Step 3: Fillet the edges; students behind finish shell and columns (28 min) Step 4: Read your timeline: find one feature and edit it instead of starting over (5 min) Step 5: Save (Ctrl+S) (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: glass, aluminum or plastic for a soda bottle - pick one and defend it. (10 min) Step 3: Finish anything left; file named Day 1 - Toy Block - Lastname in your own folder (25 min) Step 4: Check-in: show me the model, say what shell did, show one timeline edit (8 min) Step 5: Save (Ctrl+S); checked-off students help a neighbor (2 min)	Step 1: Bell ringer / essential question. Step 2: Finish and save whatever you owe me; everyone else starts the vocabulary sheet (15 min) Step 3: CAD vocabulary in your own words (8 min) Step 4: Turn-in check: both files in Students > your name, sheet handed in (2 min)
Resources: <i>Textbook, handouts, visuals, laptops.</i>				
Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.				
Re-Teaching Activities: <i>What you prepare if students do not master the objective.</i>				
Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.				
Reflection: <i>Students analyze what they accomplished and how they reached mastery.</i>				
Exit ticket – students state what they accomplished and what worked best.				
Study Skills: <i>Skills implemented by students – note-taking, summarizing, teamwork, etc.</i>				
Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.				
SPED & 504 Accommodations / Modifications: <i>Codes.</i>				
Per Individualized Education Plan (IEP) / 504 plan.				
ELL Accommodations: <i>Codes.</i>				
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).				